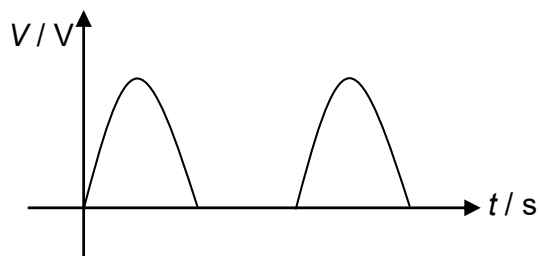


26 The figure belows shows a varying voltage.



The r.m.s. value of this rectified voltage is 100 V. The peak voltage is

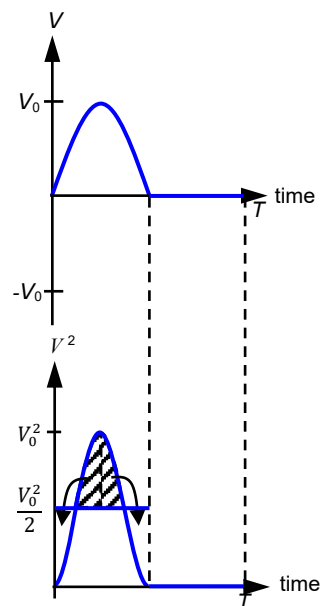
- A** 50 V **B** 71 V **C** 141 V **D** 200 V

Ans: D

$$\text{Area under } V^2\text{-}t \text{ graph for one period} = \frac{V_0^2}{2} \times \frac{T}{2} = \frac{V_0^2 T}{4}$$

$$\text{RMS} = \sqrt{\frac{\left(\frac{V_0^2 T}{4}\right)}{T}} = \frac{V_0}{2} = 100$$

$$\text{So } V_0 = 200 \text{ V}$$



27 The figure shows the variation of the photoelectric current i with voltage V between the